

Query Image + Question

Q: In which country is this building located?

Stage 1: Coarse

CLIP + FAISS
(OMGM's Pretrained Tools)

image → entity summaries
Top-20 candidates

Stage 2: Hybrid

Q-Former Reranker
(OMGM's Pretrained Tools)

(image + question) vs (image + sections) → Top-1

score gap = s1 − s2=Δ < 0.025 ?

No

Stage 3: Fine

Section Selector (BGE Cross Encoder)

text reranker over top1 entity sections

Answer Generation

Llava-1.5-7B + Llava-1.5-13b
(Fine-tuned on infoseek queries and oven images)

Answer: France

Yes

Rule-Based Agent Controller

1. confidence = score gap Δ = s1 − s2 (top-1 vs top-2); threshold 0.025
2. bounded retry budget
3. early exit when confident

Agentic Verification & Routing

1. Debiased Symmetric Visual Choice
2. Category Compatibility Check
3. BGE score on top 2 Entity

Dynamic Entity Switch?

Yes

No

Candidate 2+ section 2 selected

Candidate 1 Selected